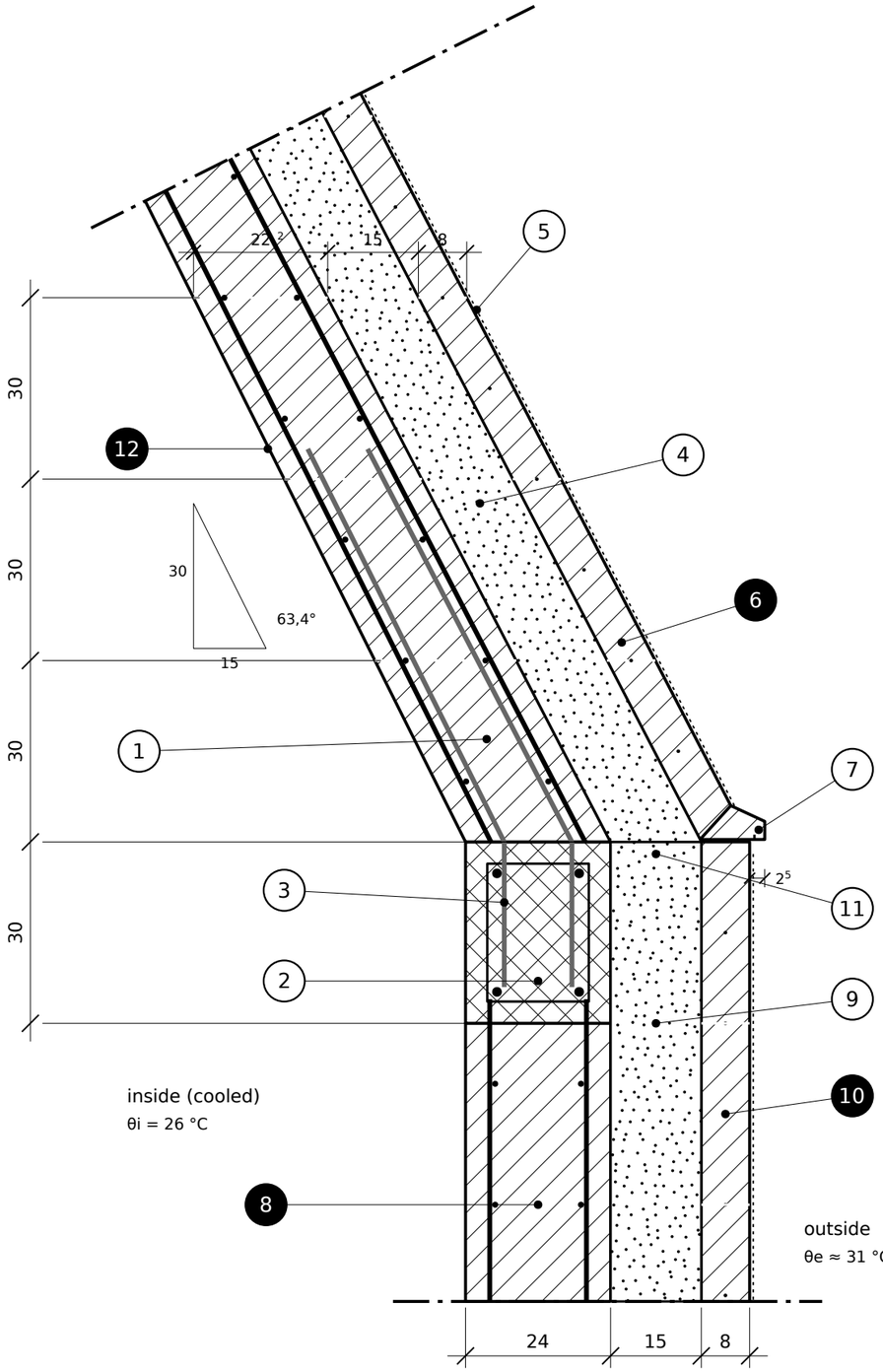
Eave section, roof-to-wall junction

Printed pieces: inner leaf / filled lattice / skin · Ishigaki · 1:10 at A3 · cm



1. Roof inner leaf, printed reinforced concrete, 24 at the ring beam tapering to 8 (one bead) at the top of the 8 m roof; bars in printed holes stop at 9 (z ≈ 7.5 m)
2. Ring beam 24 × 30 on the wall inner leaf, cast in place, 4 Ø16, stirrups Ø8 / 150; ties the roof's outward push
3. Starter bars Ø12 from the ring beam into the roof leaf, lap 650
4. Printed lattice 150 (your idea): one 20 mm zig-zag bead at ≈ 20°, touching each leaf every ≈ 80 cm (see plan); cells filled with cork granulate or water-repellent perlite (λ ≈ 0.05); U ≈ 0.47
5. Breathable water-repellent mineral coating on all outside faces (moisture control on the outside in a cooled building)
6. Roof outer skin 80, printed with the lattice and inner leaf as one piece; stainless wire in bed joints; light colour
7. Printed drip nose 2^5 on the bottom roof course, so rain drips clear of the wall; no gutter
8. Wall inner leaf, reinforced concrete 240, mesh both faces
9. Wall lattice 150, same as (4)
10. Wall outer skin 80, same as (6)
11. Fill runs on from the wall lattice into the roof lattice: no gap in the insulation at the eave; cap it at the top opening
12. Room-side face: printed concrete, pitch 63.4° (course 300, step 150 from the roof script)

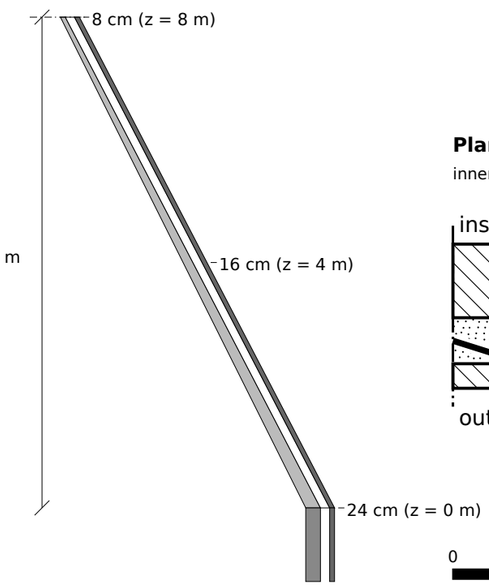
● from scripts/cork_house_roof.py or given by you
○ assumed by me: not defined yet

Notes

Each piece = inner leaf + lattice + outer skin, printed in one go, open top/bottom.
Ribs are the heat bridges: few, long, flat ribs (≈ 20°) keep U ≈ 0.47. Empty cells ≈ 2.0.
Inner leaf 24 → 8 (load ≈ 0.3 MPa at the base). Widths horizontal, as printed.
Hand estimates, not engineered or simulated: confirm structure + U-value.

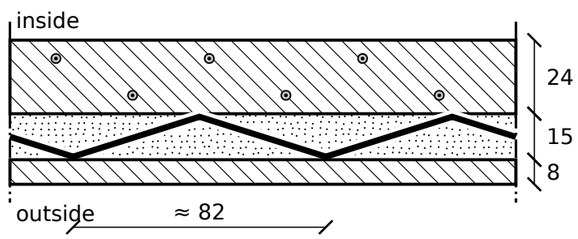
Whole roof, 1:100

inner leaf (grey) tapers 24 → 8



Plan of one wall piece, 1:20

inner leaf with rebar holes · zig-zag rib, cells filled · outer skin



0 10 20 30 40 50 cm